

One Mathematics of Arrays from Transformer Semantics to Machine Execution

Shape, Indexing, Normal Forms, Machine Translation, Prediction, and Cross-Validation

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Abstract

This working document unifies the current Mathematics of Arrays (MoA) Transformer derivation with the larger experimental and compiler program. The governing claim is deliberately simple: there is one array mathematics from semantic data to shaped machines. Arrays are defined by shape; valid index spaces are generated from shape; components are selected by ψ ; and computation is reduced to a machine-independent Denotational Normal Form (DNF). Only after the DNF is fixed is a realization introduced through reshape/dimension lifting, a chosen member of the γ family, and rav . Row-major γ is used as the default concrete example, but column-major and other layout functions are equally definable. A change from one γ to another is an explicit translation and its cost must be derived rather than hidden.

The same theory is required to explain and predict the independent CPU, GPU, multi-GPU, distributed, and heterogeneous CPU–GPU experiments. Those experiments are not separate performance models. They are independent tests of the same shape/index/mapping theory and therefore must mutually corroborate its semantic, resource, communication, and predictive consequences. The Transformer is developed under the same discipline. The existing MoA attention-family derivations are treated as established components and are composed with normalization, residual, and feed-forward definitions to obtain a complete block DNF before any machine is introduced.

1 One Theory, All the Way

The governing research chain is

FORMULATE $\rightarrow$ DERIVE/OPTIMIZE $\rightarrow$ DNF
 $\xrightarrow{\gamma}$ SEQUENTIAL ONF $\rightarrow$ LIFTED ONF
 $\rightarrow$ MACHINE ARRAY $\rightarrow$ PREDICT/FREEZE $\rightarrow$ EXECUTE/VALIDATE.

The central invariant is

The DNF never changes when the target machine changes.

The algorithm, its data, its dependency structure, and the target machine are all viewed as arrays. The same MoA rules apply on both sides of the conventional hardware/software boundary. The backend may be CPU, GPU, multi-GPU, MPI+OpenMP, OpenACC, CUDA, or concurrent CPU–GPU; the mathematical semantics are not replaced by a backend-specific theory.

1.1 The minimal semantic vocabulary

For an array A , its shape is

$$\boxed{\rho A.}$$

Its valid multidimensional index domain is generated from that shape:

$$\boxed{\iota(\rho A).}$$

For every valid index $\vec{i} \in \iota(\rho A)$, the corresponding component is

$$\boxed{\vec{i}\psi A.}$$

The complete valid index structure reconstructs the array:

$$\boxed{(\iota(\rho A))\psi A \equiv A.}$$

The canonical catenation operator is $\#$. Dimensionality is scalar:

$$\boxed{\delta(A) = \tau(\rho A).}$$

These definitions do not depend on whether A represents data, a dependency object, a partition, a processor array, a cache hierarchy, a set of GPUs, a collection of nodes, or another machine property.

1.2 The machine is an array too

Let the target machine be M , with

$$\boxed{\rho M = \langle m_0 \ m_1 \ \cdots \ m_{r-1} \rangle.}$$

Then

$$\iota(\rho M)$$

is its shaped machine-index space, and a valid machine component is selected by the same operation

$$\vec{p}\psi M, \quad \vec{p} \in \iota(\rho M).$$

Thus the common foundation is

DATA ARRAY ρ, ι, ψ	$\longleftrightarrow$	COMPUTATION ARRAY ρ, ι, ψ	$\longleftrightarrow$	MACHINE ARRAY $\rho, \iota, \psi.$
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2 DNF, Sequential ONF, and the γ Family

The DNF contains the architecture-independent index/value computation. No concrete machine address belongs in the DNF. The first operational realization is always the one-to-one sequential ONF:

$$\boxed{\text{DNF}} \xrightarrow[\text{one-to-one}]{\gamma} \boxed{\text{Sequential ONF}}.$$

2.1 First use of γ : storage-order convention

The function γ translates a multidimensional index, together with an array shape, to a linear address in a raveled representation. Unless otherwise stated, concrete examples in this document use a **row-major** γ . This is a realization choice, not a restriction of MoA. A column-major γ is also defined, and other γ 's may be designed for tiled, blocked, strided, sparse, hierarchical, or architecture-specific layouts.

For

$$\rho A = \langle n \ m \rangle,$$

the row-major member is

$$\boxed{\gamma_{\text{rm}}(\langle i \# j \rangle; \rho A) = (i \times m) + j,}$$

and

$$\boxed{\langle i \# j \rangle \psi A \longrightarrow (\text{rav } A)[(i \times m) + j].}$$

For the same abstract array, a column-major member is

$$\boxed{\gamma_{\text{cm}}(\langle i \# j \rangle; \rho A) = i + (j \times n).}$$

Hence the DNF may have multiple legal operational realizations:

$$\boxed{\text{DNF}(F)} \xrightarrow{\gamma_{\text{rm}}} \text{ONF}_{\text{rm}}(F), \quad \boxed{\text{DNF}(F)} \xrightarrow{\gamma_{\text{cm}}} \text{ONF}_{\text{cm}}(F).$$

2.2 Changing γ has a cost

If one stage is realized under γ_a and a later stage requires γ_b , the change is not silently free. Either the new access can be interpreted without physical reorganization, or an explicit translation is required:

$$\boxed{\gamma_a \longrightarrow T_{a \rightarrow b} \longrightarrow \gamma_b.}$$

The translation has a deterministic resource consequence

$$\boxed{C(T_{a \rightarrow b})}$$

that must be included in the mapped cost. Thus layout becomes a mathematical design variable. A predictive realization may compare

$$\gamma_1, \gamma_2, \dots, \gamma_k \longrightarrow C_1, C_2, \dots, C_k$$

before execution.

3 Reshape and Dimension Lifting

Dimension lifting occurs *after* the sequential ONF has been formed and verified. It does not alter the DNF.

If a sequential execution index space has shape ρP , a legal lifted shape ρP^L must preserve the number of represented components:

$$\boxed{\tau(\rho P^L) = \tau(\rho P).}$$

A canonical lifting pattern is

$$\boxed{N \rightarrow \Pi \# (N \div \Pi),}$$

with the lifted application

$$\boxed{(f\Omega^{(1)})((\Pi \# (N \div \Pi))\rho N).}$$

The semantic invariant is

$$\boxed{\vec{i}\psi Output_{\text{DNF}} = \vec{i}\psi Output_{\text{seq}} = \vec{i}\psi Output_{\text{lifted}}}.$$

Therefore

Dimension lifting changes execution shape, not semantic meaning.

The lifted shape also supplies the legal loop/index bounds. For

$$\rho P^L = \langle p_0 \ p_1 \ \cdots \ p_{k-1} \rangle,$$

the corresponding bounds are

$$0 \leq i_0 < p_0, \dots, 0 \leq i_{k-1} < p_{k-1}.$$

4 Prediction and Independent Experimental Cross-Validation

The experimental discipline is

derive $\rightarrow$ map $\rightarrow$ predict $\rightarrow$ **FREEZE** $\rightarrow$ first execution $\rightarrow$ validate.

The independent experiments must not become unrelated empirical models. Each experiment instantiates the same MoA theory with a different computation shape and/or machine shape. Consequently, the experiments cross-validate the theory by testing different consequences of the same definitions.

The separation is:

Exact semantic quantities	shapes, valid indices, dependencies, work, ownership, bytes, communication
Machine realization properties	capacities, rates, paths, latency, bandwidth, realization factors
Validation quantities	correctness, first execution, measured traffic/time, prediction error

No target measurement used for validation may be retroactively inserted into the frozen prediction.

4.1 What the CPU experiments teach

The multicore CPU experiments instantiate shaped processors, cache domains, NUMA domains, and memory resources. The same computation DNF produces different realized behavior when its ONF is mapped to different machine shapes. Controlled compact-versus-balanced NUMA experiments show that worker count alone is insufficient: the actual indexed machine shape and first-touch placement matter.

At 16384², representative 1-to-64-core measurements include:

Program	T_1 (s)	T_{64} (s)	Speedup	Efficiency
Sobel	1.178684	0.018703	63.02	98.5%
Reduce	0.366707	0.005836	62.84	98.2%
Histogram	0.157111	0.002637	59.59	93.1%
Pipeline	0.844528	0.015650	53.96	84.3%
Otsu	0.186954	0.003935	47.51	74.2%
RGB2Gray	0.105370	0.002781	37.89	59.2%
Threshold	0.030802	0.001728	17.82	27.8%

The different scaling curves are consequences to be explained from different ψ -derived dependency structures under the same machine.

4.2 What the GPU and multi-GPU experiments teach

GPU experiments retain the semantic DNF while replacing the machine array and operational mapping. Pointwise work, neighborhood work, reductions, shape-changing remaps, and composed pipelines therefore become independent tests of the same theory.

For a row-partitioned pointwise array,

$$\rho A = \langle H \ W \rangle, \quad \rho A_p = \langle H/P \ W \rangle,$$

aligned ownership gives no semantic halo:

$$H_p = \emptyset.$$

For Sobel, the 3×3 dependency causes required indices to cross an ownership boundary. For a two-way row split the exact overlap is two rows:

$$N_{\text{halo}} = 2W.$$

The distinction is derived from indexing, not from a GPU-specific communication rule.

4.3 What heterogeneous CPU–GPU experiments teach

Concurrent CPU–GPU execution tests a stronger consequence: the same semantic index space may be divided across unlike machine components. The mapped completion time must account for the distinct machine realization functions and any communication implied by the dependency structure.

For the pointwise Threshold experiment with $N = 2^{30}$, the frozen heterogeneous split was

$$N_C = 689,968,322, \quad N_G = 383,773,502,$$

with

$$\hat{T}_{\text{Threshold}} = 86.2055821 \text{ ms.}$$

The first measured completion was

$$T_{\text{measured}} = 85.816671 \text{ ms,}$$

giving

$$\text{prospective error} = -0.451\%.$$

A separately frozen heterogeneous Halide Sobel mapping predicted

$$\hat{T}_{\text{Sobel}} = 6.495540 \text{ ms}$$

and measured

$$T_{\text{measured}} = 6.465319 \text{ ms,}$$

for

$$\text{prospective error} = -0.4653\%.$$

The Sobel case additionally exposed a critical MoA point: nominal CPU identifiers did not initially represent the intended physical-core machine shape. Correcting the *machine mapping*, while leaving the frozen semantic design unchanged, restored correspondence between prediction and execution.

These experiments therefore validate different parts of one chain:

$$\text{indexing} \rightarrow \text{dependency} \rightarrow \text{ownership} \rightarrow \text{communication} \rightarrow \text{machine mapping} \rightarrow \text{prediction}.$$

4.4 Composition validates a resource tradeoff, not automatic speedup

The composed RGB2Gray–Reduce–Sobel work reduced principal storage approximately from

$$M_{\text{separate}} \approx 11N \quad \text{to} \quad M_{\text{composed}} \approx 7N.$$

The largest successful square side increased from 87,552 to 109,824, corresponding to a 57.3% increase in feasible pixel count. At 2048², however, the composed kernel was about 24% slower because grayscale values were recomputed. This is exactly the kind of tradeoff the theory must expose: storage reduction is a provable semantic/resource consequence; execution time follows from the selected realization and machine properties.

5 The Existing Attention Family Is Part of the Same Theory

The attention papers are not external kernels to be wrapped by a new Transformer theory. They are prior MoA derivations to be composed.

The established progression is

$$\begin{aligned} &\text{forward attention} \rightarrow \text{backward/fused training} \\ &\quad \rightarrow \text{decode/KV cache/GQA/MQA} \rightarrow \text{complete pre-norm block.} \end{aligned}$$

Forward attention uses systematic ψ reduction to access Q, K, V directly, eliminating an explicit K^T buffer and unnecessary score/softmax intermediates. Backward attention applies the same discipline to the softmax VJP and gradients. Decode specializes the DNF to a current query row, expresses KV-cache accumulation with canonical catenation $\#$, and expresses GQA/MQA through ψ selection. The normalization/FFN work extends the same method to RMSNorm/LayerNorm and gated MLP/SwiGLU.

Thus the Transformer work below must reuse those derivations rather than recreate conventional tensor formulas and translate them afterward.

6 Complete Transformer Block: Architecture-Independent Formula-tion

Let the input to block l be

$$\rho X^{(l-1)} = \langle B \ N \ D \rangle$$

and the output satisfy

$$\rho X^{(l)} = \langle B \ N \ D \rangle.$$

For the pre-norm block, introduce names only for expressions that will subsequently be substituted:

$$Z^{(l)} = \text{Norm}_1^{(l)}(X^{(l-1)}),$$

$$A^{(l)} = \text{Attn}^{(l)}(Z^{(l)}),$$

$$R^{(l)} = X^{(l-1)} + A^{(l)},$$

$$W^{(l)} = \text{Norm}_2^{(l)}(R^{(l)}),$$

$$F^{(l)} = \text{FFN}^{(l)}(W^{(l)}),$$

and

$$X^{(l)} = R^{(l)} + F^{(l)}.$$

These names do not assert materialized arrays. They are substitution points on the path to DNF.

6.1 Block boundary in ψ notation

For every

$$\langle b\#n\#d \rangle \in \iota(\rho X^{(l)}),$$

$$\begin{aligned} \langle b\#n\#d \rangle \psi X^{(l)} &= \langle b\#n\#d \rangle \psi X^{(l-1)} \\ &\quad + \langle b\#n\#d \rangle \psi A^{(l)} \\ &\quad + \langle b\#n\#d \rangle \psi F^{(l)}. \end{aligned}$$

This is the component from which the complete block DNF is derived.

7 Normalization Defined by Shape

To avoid collision with the fundamental translator γ , this working document denotes the learned RMSNorm scale by $\vec{g}_1^{(l)}$ and $\vec{g}_2^{(l)}$. The symbol γ is reserved for the DNF-to-ONF address/layout family.

For fixed $\langle b\#n \rangle$, define the first normalization statistic

$$r_{1,b,n}^{(l)} = \sqrt{\frac{1}{D} \sum_{e \in \iota(D)} (\langle b\#n\#e \rangle \psi X^{(l-1)})^2 + \epsilon}.$$

Then

$$\langle b\#n\#d \rangle \psi Z^{(l)} = g_{1,d}^{(l)} \frac{\langle b\#n\#d \rangle \psi X^{(l-1)}}{r_{1,b,n}^{(l)}}.$$

The residual component is

$$\langle b\#n\#d \rangle \psi R^{(l)} = \langle b\#n\#d \rangle \psi X^{(l-1)} + \langle b\#n\#d \rangle \psi A^{(l)}.$$

The second normalization statistic is therefore

$$r_{2,b,n}^{(l)} = \sqrt{\frac{1}{D} \sum_{e \in \iota(D)} (\langle b\#n\#e \rangle \psi R^{(l)})^2 + \epsilon}.$$

and

$$\langle b\#n\#d \rangle \psi W^{(l)} = g_{2,d}^{(l)} \frac{\langle b\#n\#d \rangle \psi R^{(l)}}{r_{2,b,n}^{(l)}}.$$

All bounds arise from shapes. No machine dimension has been introduced.

8 FFN Defined by Shape and ψ

Let

$$\rho W_{\text{gate}}^{(l)} = \rho W_{\text{up}}^{(l)} = \langle D \ D_{\text{ff}} \rangle, \quad \rho W_{\text{down}}^{(l)} = \langle D_{\text{ff}} \ D \rangle.$$

For

$$f \in \iota(D_{\text{ff}}),$$

define

$$u_{b,n,f}^{(l)} = \sum_{d' \in \iota(D)} (\langle b\#n\#d' \rangle \psi W^{(l)}) \times (\langle d' \# f \rangle \psi W_{\text{gate}}^{(l)}),$$

$$v_{b,n,f}^{(l)} = \sum_{d' \in \iota(D)} (\langle b \# n \# d' \rangle \psi W^{(l)}) \times (\langle d' \# f \rangle \psi W_{\text{up}}^{(l)}),$$

and

$$h_{b,n,f}^{(l)} = \text{SiLU}(u_{b,n,f}^{(l)}) \times v_{b,n,f}^{(l)}.$$

Then

$$\langle b \# n \# d \rangle \psi F^{(l)} = \sum_{f \in \iota(D_{\text{ff}})} h_{b,n,f}^{(l)} \times (\langle f \# d \rangle \psi W_{\text{down}}^{(l)}).$$

Substitution removes U, V, H as required semantic intermediates:

$$\begin{aligned} \langle b \# n \# d \rangle \psi F^{(l)} = \sum_{f \in \iota(D_{\text{ff}})} & \left[\text{SiLU} \left(\sum_{d' \in \iota(D)} (\langle b \# n \# d' \rangle \psi W^{(l)}) (\langle d' \# f \rangle \psi W_{\text{gate}}^{(l)}) \right) \right. \\ & \times \left(\sum_{d' \in \iota(D)} (\langle b \# n \# d' \rangle \psi W^{(l)}) (\langle d' \# f \rangle \psi W_{\text{up}}^{(l)}) \right) \\ & \left. \times (\langle f \# d \rangle \psi W_{\text{down}}^{(l)}) \right]. \end{aligned}$$

The remaining $W^{(l)}$ occurrences are then replaced by the second-normalization component definition, and $R^{(l)}$ is replaced by its residual component definition.

9 Attention Is Substituted from the Established MoA DNF

The next step is not to invent another attention formula. Every occurrence of $Z^{(l)}$ in the established attention-family DNF is replaced by the first-normalization ψ expression above. The attention result $A^{(l)}$ is then substituted into both the residual and the FFN path.

The derivation therefore proceeds

$$\begin{aligned} \text{Transformer formulation} & \rightarrow \text{established MoA attention DNF} \rightarrow \text{residual/norm substitution} \\ & \rightarrow \text{FFN substitution} \rightarrow \psi \text{ reduction} \rightarrow \text{candidate block DNF.} \end{aligned}$$

The candidate DNF is frozen only after component, shape, and numerical verification.

10 Transformer Invariants and Verification Gate

The first block invariant is

$$\rho X^{(l)} = \rho X^{(l-1)} = \langle B \ N \ D \rangle.$$

Thus blocks compose:

$$X^{(0)} \xrightarrow{B^{(1)}} X^{(1)} \xrightarrow{B^{(2)}} \dots \xrightarrow{B^{(L)}} X^{(L)},$$

with

$$\rho X^{(0)} = \rho X^{(1)} = \dots = \rho X^{(L)} = \langle B \ N \ D \rangle.$$

At every transformation boundary the verification rule is

$$\text{same semantic function} + \text{same result shape} + \text{same components.}$$

After dimension lifting, add the execution-space invariant

$$\tau(\rho P^L) = \tau(\rho P).$$

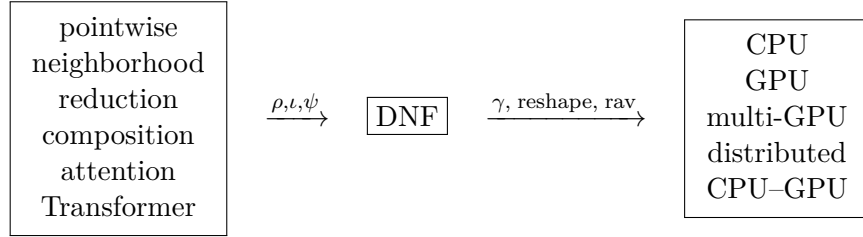
Therefore the complete gate is

$$\text{derive} \rightarrow \text{verify semantics} \rightarrow \text{verify shape} \rightarrow \text{verify components} \rightarrow \text{FREEZE} \rightarrow \text{continue}.$$

Only after the complete Transformer-block DNF passes this gate do we introduce a chosen γ , sequential ONF, dimension lifting, and a target machine array.

11 The Unified Scientific Test

The CPU, GPU, multi-GPU, distributed, CPU–GPU, attention, decode, and Transformer experiments must validate one another in the following precise sense: each is an independent instantiation of the same theory and therefore tests a different implication of the same invariant definitions.



A contradiction between independently derived experiments is therefore scientifically useful: it identifies either an incorrect semantic derivation, an incorrect machine description, an incorrect mapping, or an incomplete machine-cost property. It must not be repaired by changing the theory independently for each backend.

Conversely, agreement across independently frozen experiments strengthens the common theory because the same definitions survive changes in algorithm class, dimensionality, machine shape, and realization.

12 Next Mathematical Step

The immediate next step is to complete the substitution

$$\langle b \# n \# d \rangle \psi X^{(l)}$$

using the established MoA attention-family DNF until only true block inputs, weights, scalar operations, reductions, ρ , ι , and ψ remain.

At that point:

$$\text{complete Transformer-block DNF}$$

is verified and frozen.

Only afterward:

$$\begin{array}{c} \text{DNF} \xrightarrow{\gamma_{\text{chosen}}} \text{Sequential ONF} \xrightarrow{\text{verify}} \text{Dimension Lift} \\ \xrightarrow{\text{map}} \text{Machine Array} \xrightarrow{\text{predict/freeze}} \text{execute/validate.} \end{array}$$

The same procedure then scales from one block to multiple blocks and eventually to complete models without changing the underlying mathematics.

Working Principle

Define by shape. Generate indices with ι . Select with ψ .

Reduce to one invariant DNF.

Translate with an explicit γ , reshape/lift, and rav .

Treat the machine as an array. Predict, freeze, execute, validate.

One theory. Independent experiments. Mutual validation. MoA all the way.

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